

Isaac Li | Curriculum Vitae

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Education

University of Southern California

Ph.D. Computer Science

Los Angeles, California

August 2026–Present

- Operations Research and Artificial Intelligence Fellow (NSF Research Traineeship)
- Advised by Evi Micha

University of Rochester

B.S. Mathematics, Honors with Highest Distinction

Honors Thesis: [Annihilating Pairs Inequalities in Orlicz Spaces](#)

Rochester, New York

August 2021–May 2025

B.A. Economics, Honors with High Distinction

Honors Thesis: Semiconductor Shortage, Computer Prices, and Investment in Tech Industries

B.S. Computer Science, High Distinction

- GPA: 3.81/4.00 (Cum Laude, Dean's List)
- Certificate of Achievement in Mathematical Modeling in Political Science and Economics
- Citation of Special Achievement in Economics
- Rush Rhees Scholarship, National Merit Scholar

Articles

In Press

- [1] M. Dannenberg, W. Hagerstrom, G. Hart, A. Iosevich, T. Le, **I. Li**, and N. Skerrett. *Buffon Needle Problem Over Convex Sets*, 2024. Accepted in the **Fractal Geometry and Harmonic Analysis** collection of **Research in the Mathematical Sciences**. [arXiv:2411.16935](#).
- [2] A. Iosevich, **I. Li**, Z. Li, and E. Yu. *Orlicz Spaces and the Uncertainty Principle*, 2025. Accepted in **Analysis Mathematica**. [arXiv:2509.05185](#).

Preprints

- [3] K. Aldaleh, W. Burstein, G. Garza, G. Hart, A. Iosevich, J. Iosevich, A. Khalil, J. King, N. Kulkarni, T. Le, **I. Li**, A. Mayeli, B. McDonald, K. Nguyen, and N. Shaffer. *The Fourier Ratio and Complexity of Signals*, 2025. [arXiv:2511.19560](#).

Experience

Research and Teaching

StemForAll

Program Instructor

Rochester, New York

July 2025–August 2025

- Investigated the relationship between the learnability of a signal with its “Fourier Ratio,” the ratio between the L^1 and L^2 norms of its Fourier transform
- Proved that signals with low Fourier Ratio allow approximation via low-degree trigonometric polynomials, have low algorithmic rate-distortion and VC dimension, and display additive structure in high valued sets
- Wrote Python simulations to numerically estimate constants in Bourgain- and Talagrand-type inequalities

University of Rochester Department of Computer Science

Teaching Assistant, Design and Analysis of Efficient Algorithms

Rochester, New York

January 2024–May 2024

- Held recitation sessions and graded homeworks and exams covering topics including greedy algorithms, linear programming, dynamic programming, graph algorithms, divide and conquer algorithms, and P vs NP

NSF Tripods REU

Researcher

Rochester, New York

July 2023–August 2023

- Investigated the behavior of needles falling randomly on sets, similar to the classical Buffon needle problem
- Proved that for any convex set with given perimeter, there is a sufficiently small needle for which the circle maximizes the “Buffon Probability” that the needle lands entirely within the set

- Derived explicit formulas for the Buffon probability over circles, ellipses, and regular polygons
- Wrote Python simulations to numerically confirm theoretical results

Health and Environmental Economics Lab

Research Assistant

Rochester, New York

May 2022–August 2022

- Utilized parallel processing to significantly improve runtime of OCR data processing script
- Completed dataset which previously took 3 months in 23 hours

Professional

Esperanto Technologies

Machine Learning Intern

Mountain View, California

May 2024–December 2024

- Investigated the theory behind homomorphic encryption techniques (primarily CKKS) to evaluate feasibility for implementation and optimization on Esperanto hardware
- Studied approximate k -nearest neighbor algorithms for use in retrieval augmented generation

Machine Learning Intern

May 2023–August 2023

- Implemented switch transformers on Vicuna by modifying the Llama architecture to add an MoE layer
- Analyzed RoBERTa parameter and activation distributions to investigate opportunities for quantization

Skills

Languages: C/C++, Java, Python

Libraries: PyTorch, NumPy, SciPy

Communication: L^AT_EX, Word, Powerpoint